

DUMAGUETE, Philippines

WMO: 986420

Lat: 9.334N

Lon: 123.300E

Elev: 15

StdP: 14.69

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	73.5	74.4	69.6	108.7	76.8	70.6	112.9	77.3	22.0	79.6	18.1	80.3	1.3	320	0.347

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	11.3	91.9	81.0	91.0	81.0	90.0	80.8	82.7	89.5	82.1	89.0	81.6	88.4	8.6	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.8	160.2	87.8	80.1	156.4	87.3	79.5	153.1	86.8	46.7	89.7	46.0	89.3	45.3	89.0	89.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.1	13.5	11.9	DB	67.5	96.3	4.5	4.0	64.3	99.2	61.7	101.6	59.1	103.8	55.9	106.7	
			WB	66.1	84.7	4.1	1.6	63.1	85.8	60.7	86.7	58.4	87.6	55.4	88.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	82.0	80.3	80.4	81.4	82.8	83.6	82.9	82.1	82.5	82.3	82.0	82.1	81.5
	DBStd	1.89	1.62	1.60	1.52	1.55	1.68	1.77	1.68	1.90	1.61	1.63	1.54	1.52
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11685	939	852	973	985	1042	988	996	1008	969	993	962	978
	CDD65	6210	474	432	508	535	577	538	531	543	519	528	512	513
	CDH74	68323	4476	4219	5462	6419	7114	6278	5819	6156	5650	5691	5640	5399
	CDH80	23126	886	961	1629	2431	2975	2466	2080	2397	2036	1957	1833	1475
Wind	WSAvg	4.6	5.9	5.7	5.2	4.7	4.2	3.8	3.7	4.3	3.8	3.9	4.5	5.3
Precipitation	PrecAvg	47.8	3.4	2.6	2.6	1.5	3.3	4.1	5.6	4.4	5.2	6.2	6.1	3.9
	PrecMax	74.1	11.2	10.3	8.3	5.4	7.3	8.6	12.1	10.3	20.5	13.9	11.1	7.2
	PrecMin	26.5	0.2	0.5	0.1	0.2	0.4	1.3	0.5	1.9	1.7	1.7	2.8	1.4
	PrecStd	12.4	3.0	2.5	2.7	1.3	2.0	2.0	3.3	2.1	4.5	4.1	2.8	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	87.6	87.8	89.1	90.9	93.0	93.3	92.7	93.3	92.4	91.2	89.9	89.1
		MCWB	79.4	79.3	79.8	81.3	81.2	80.8	80.9	80.8	81.1	81.0	80.5	80.7
	2%	DB	86.2	86.4	88.0	89.5	91.0	91.4	90.6	91.6	90.9	89.6	88.6	87.9
		MCWB	79.1	78.7	79.3	80.5	81.4	81.2	80.6	80.7	80.8	80.5	80.4	80.3
	5%	DB	85.3	85.8	87.0	88.5	89.9	89.9	89.3	90.0	89.6	88.5	87.8	86.8
		MCWB	78.8	78.4	79.0	80.2	81.1	80.9	80.5	80.6	80.6	80.4	80.3	79.8
	10%	DB	84.2	84.8	86.1	87.8	89.1	88.8	88.0	88.9	88.2	87.6	86.9	85.8
		MCWB	78.3	78.1	78.7	79.9	80.9	80.6	80.2	80.3	80.2	80.2	79.9	79.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.7	80.6	81.2	82.6	83.7	83.2	82.6	83.4	83.0	82.5	81.9	81.9
		MCDB	85.7	86.2	87.4	89.4	89.8	90.2	89.6	90.4	89.7	88.8	87.6	87.2
	2%	WB	79.8	79.7	80.2	81.4	82.5	82.1	81.5	81.9	81.8	81.5	81.0	80.8
		MCDB	84.8	85.2	86.3	88.2	89.4	89.4	88.7	89.4	89.1	87.9	87.0	86.4
	5%	WB	79.1	79.0	79.7	80.8	81.8	81.5	80.8	81.0	81.0	80.8	80.6	80.2
		MCDB	84.1	84.5	85.7	87.4	88.7	88.6	88.0	88.6	88.2	87.4	86.7	85.6
	10%	WB	78.5	78.5	79.1	80.2	81.2	80.8	80.2	80.4	80.3	80.2	80.1	79.6
		MCDB	83.3	83.8	85.0	86.6	88.0	87.8	87.0	87.9	87.3	86.6	85.9	84.8
Mean Daily Temperature Range	5% DB	MDBR	7.7	8.5	9.6	10.0	11.3	11.8	11.9	12.9	12.2	11.3	10.0	8.6
		MCDBR	9.2	10.1	11.0	10.9	12.5	13.5	14.0	14.9	14.3	12.9	11.0	10.1
		MCWBR	4.9	5.3	5.7	5.5	6.5	6.7	6.8	7.3	7.2	6.6	5.7	5.3
	5% WB	MCDBR	8.5	9.1	9.9	10.4	11.9	12.7	13.1	14.0	13.4	12.3	10.6	9.6
		MCWBR	4.8	5.1	5.5	5.5	6.6	6.8	6.8	7.4	7.2	6.5	5.6	5.2
Clear-Sky Solar Irradiance	taub	0.423	0.418	0.421	0.432	0.431	0.421	0.421	0.440	0.436	0.448	0.430	0.426	
	taud	2.455	2.456	2.451	2.419	2.433	2.475	2.477	2.403	2.408	2.372	2.437	2.447	
	Ebn at Noon	282	287	286	279	274	273	274	273	277	274	278	278	
	Edh at Noon	36	37	37	38	37	35	35	38	38	39	36	36	
All-Sky Solar Radiation	RadAvg	1368	1613	1846	1968	1777	1612	1535	1587	1554	1535	1500	1383	
	RadStd	147	158	146	80	119	87	99	118	141	103	115	114	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.40	N/A	N/A	+1.10	+0.85	+0.70	N/A	N/A	N/A	N/A
(47)	Regional (7 neighbors)	+0.27	N/A	N/A	+0.72	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page